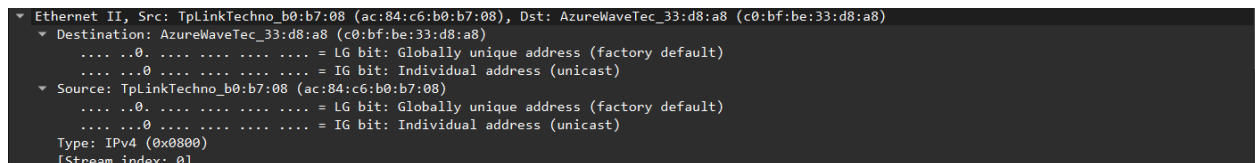
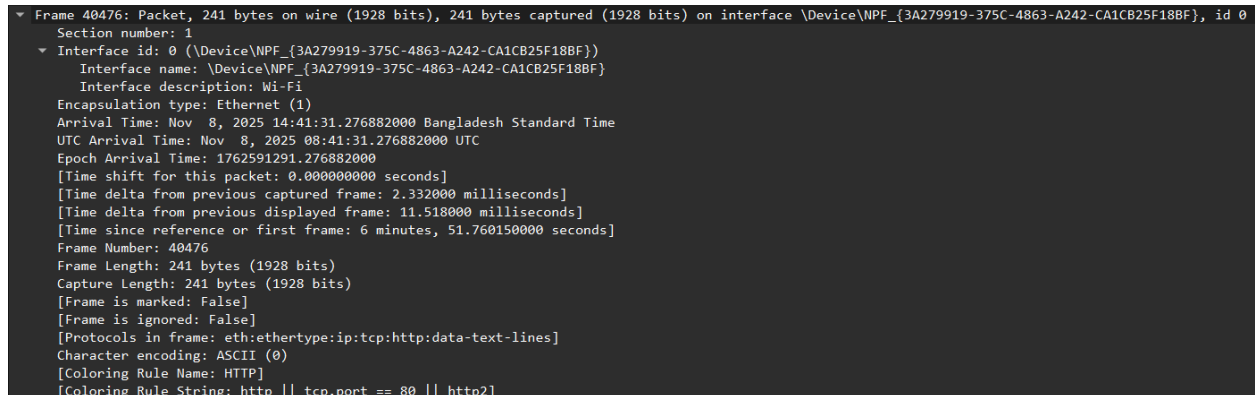
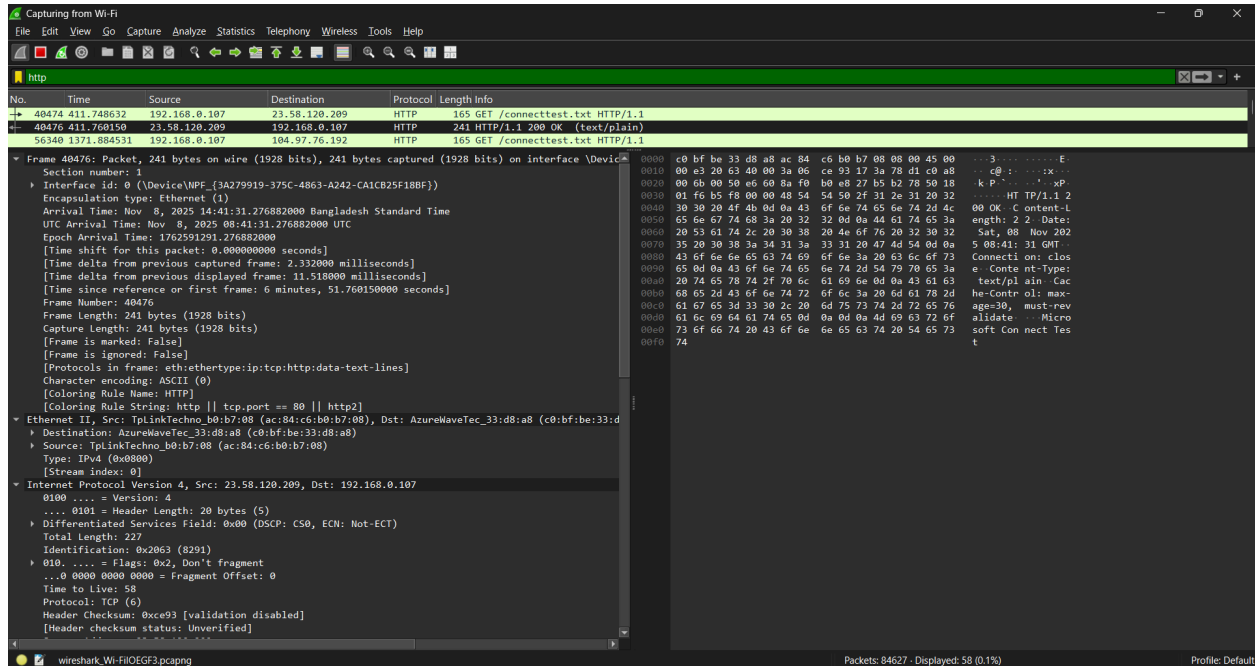


5 layers 'protocols' headers for a request packet and a response packet



```
▼ Internet Protocol Version 4, Src: 23.58.120.209, Dst: 192.168.0.107
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ▼ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    0000 00.. = Differentiated Services Codepoint: Default (0)
    .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 227
  Identification: 0x2063 (8291)
  ▼ 010. .... = Flags: 0x2, Don't fragment
    0... .... = Reserved bit: Not set
    .1.. .... = Don't fragment: Set
    ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 58
  Protocol: TCP (6)
  Header Checksum: 0xce93 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 23.58.120.209
  Destination Address: 192.168.0.107
  [Stream index: 69]
```

```
▼ Transmission Control Protocol, Src Port: 80, Dst Port: 58976, Seq: 1, Ack: 112, Len: 187
  Source Port: 80
  Destination Port: 58976
  [Stream index: 56]
  [Stream Packet Number: 6]
  ▼ [Conversation completeness: Complete, WITH_DATA (31)]
    ..0. .... = RST: Absent
    ....1 .... = FIN: Present
    .... 1... = Data: Present
    .... .1.. = ACK: Present
    .... ..1. = SYN-ACK: Present
    .... ...1 = SYN: Present
    [Completeness Flags: ·FDASS]
  [TCP Segment Len: 187]
  Sequence Number: 1 (relative sequence number)
  Sequence Number (raw): 2331029736
  [Next Sequence Number: 188 (relative sequence number)]
  Acknowledgment Number: 112 (relative ack number)
  Acknowledgment number (raw): 666219128
  0101 .... = Header Length: 20 bytes (5)
  ▸ Flags: 0x018 (PSH, ACK)
  Window: 502
  [Calculated window size: 64256]
  [Window size scaling factor: 128]
  Checksum: 0xb5f8 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  ▼ [Timestamps]
    [Time since first frame in this TCP stream: 21.889000 milliseconds]
    [Time since previous frame in this TCP stream: 2.332000 milliseconds]
  ▼ [SEQ/ACK analysis]
    [iRTT: 10.181000 milliseconds]
    [Bytes in flight: 187]
    [Bytes sent since last PSH flag: 187]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
  TCP payload (187 bytes)
```

```

Hypertext Transfer Protocol
  HTTP/1.1 200 OK\r\n
    Response Version: HTTP/1.1
    Status Code: 200
    [Status Code Description: OK]
    Response Phrase: OK
  Content-Length: 22\r\n
    [Content length: 22]
  Date: Sat, 08 Nov 2025 08:41:31 GMT\r\n
  Connection: close\r\n
  Content-Type: text/plain\r\n
  Cache-Control: max-age=30, must-revalidate\r\n
  \r\n
  [Request in frame: 40474]
  [Time since request: 11.518000 milliseconds]
  [Request URI: /connecttest.txt]
  [Full request URI: http://www.msftconnecttest.com/connecttest.txt]
  File Data: 22 bytes
Line-based text data: text/plain (1 lines)
  Microsoft Connect Test

```

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

httpresponse

No.	Time	Source	Destination	Protocol	Length	Info
66691	12510.879161	142.250.77.131	192.168.0.107	HTTP	276	HTTP/1.1 304 Not Modified
66707	12967.107467	10.11.12.13	192.168.0.107	HTTP	463	HTTP/1.1 304 Not Modified
66741	14650.072149	42.99.140.153	192.168.0.107	HTTP	241	HTTP/1.1 200 OK (text/plain)
66809	16070.800451	192.168.0.1	192.168.0.107	SSDP	527	HTTP/1.1 200 OK
66906	16070.838800	192.168.0.1	192.168.0.107	SSDP	527	HTTP/1.1 200 OK
66908	16071.005119	42.99.140.187	192.168.0.107	HTTP	241	HTTP/1.1 200 OK (text/plain)
66909	16071.005121	42.99.140.187	192.168.0.107	HTTP	241	HTTP/1.1 200 OK (text/plain)

Frame 66888: Packet, 527 bytes on wire (4216 bits), 527 bytes captured (4216 bits) on interface \Device\NPF_{3A279919-375C-4863-A242-CA1CB25F18BF}

Section number: 1

Interface id: 0 (\Device\NPF_{3A279919-375C-4863-A242-CA1CB25F18BF})

Interface name: \Device\NPF_{3A279919-375C-4863-A242-CA1CB25F18BF}

Interface description: Wi-Fi

Encapsulation type: Ethernet (1)

Arrival Time: Nov 8, 2025 19:02:30.326193000 Bangladesh Standard Time

UTC Arrival Time: Nov 8, 2025 13:02:30.326193000 UTC

Epoch Arrival Time: 1762606950.326193000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 7.755000 milliseconds]

[Time delta from previous displayed frame: 23 minutes, 40.737312000 seconds]

[Time since reference or first frame: 4 hours, 27 minutes, 50.809461000 seconds]

Frame Number: 66888

Frame Length: 527 bytes (4216 bits)

Capture Length: 527 bytes (4216 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ip:udp:ssdp]

Character encoding: ASCII (0)

[Coloring Rule Name: UDP]

[Coloring Rule String: udp]

Ethernet II, Src: TpLinkTechno_b0:b7:08 (ac:84:c6:b0:b7:08), Dst: AzureWaveTec_33:d8:a8 (c0:bf:be:33:d8:a8)

Internet Protocol Version 4, Src: 192.168.0.1, Dst: 192.168.0.107

User Datagram Protocol, Src Port: 1900, Dst Port: 63347

Simple Service Discovery Protocol

0000 c0 bf be 33 d8 a8 ac 84 c0 b0 b7 08 00 00 45 00 ... 3E:

0010 02 01 00 00 00 00 40 11 67 2f c0 a8 00 01 c0 e8 ... @ @ / ..

0020 00 6b 07 6c f7 73 01 ed 34 13 48 54 54 50 2f 31 ... k 1 s 4 HTTP/1

0030 2e 31 20 32 30 30 20 4f 4b 0d 0a 43 41 43 48 451 200 O K CACHE

0040 2d 43 4f 4e 54 52 4f 4c 3a 20 6d 61 78 2d 61 67 ... -CONTROL : max-ag

0050 65 3d 33 30 30 0d 0a 44 41 54 45 3a 20 53 61 74 ... e=300; D ATE: Sat

0060 2c 20 30 38 20 4e 6f 76 20 32 30 32 35 20 31 33 ... , 08 Nov 2025 13

0070 3a 30 32 3a 32 39 20 47 4d 54 0d 0a 45 58 54 3a ... :02:29 G MT EXT:

0080 0d 0a 4c 4f 43 41 54 49 4f 4e 3a 20 68 74 74 70 ... :LOCATI ON: http

0090 3a 2f 2f 31 39 32 2e 31 36 38 2e 30 2e 31 3a 31 ... //192.1 68.0.111

00a0 39 30 30 2f 6f 61 74 65 64 65 73 63 2e 70 6d 6c ... 900/gate desc.xml

00b0 0d 0a 4f 50 54 3a 20 22 68 74 74 70 3a 2f 2f 73 ... :OPT: " http://s

00c0 63 68 65 6d 61 73 2e 75 70 6e 70 2e 6f 72 6f 2f ... chemas.u pnp-org/

00d0 75 70 6e 70 2f 31 2f 30 2f 22 3b 20 6e 73 3d 30 ... upnp/1.0 /?; res=0

00e0 31 0d 0a 30 31 2d 4e 4c 53 3a 20 32 36 32 33 34 ... 1 -01-M S: 26234

00f0 30 65 65 2d 31 64 64 32 2d 31 31 62 32 2d 61 32 ... 0ee-1dd2 -11b2-a2

0100 32 36 2d 63 36 64 61 31 37 39 33 31 32 32 36 0d ... 26-c6da1 7931226

0110 0a 53 45 52 56 45 52 3a 20 4c 69 6e 75 78 2f 32 ... SERVER: Linux/2

0120 2e 36 2e 33 36 2c 20 55 50 6e 50 2f 31 2e 30 2c ... 6.36; U PnP/1.0;

0130 20 50 6f 72 74 61 62 6c 65 20 53 44 4b 20 66 6f ... Portabl e SDK Fo

0140 72 20 55 50 6e 50 20 64 65 76 69 63 65 73 2f 31 ... r UPnP d evices/1

0150 2e 36 2e 31 39 0d 0a 58 2d 55 73 65 72 2d 41 676.19: X -User-Ag

0160 65 6e 74 3a 20 72 65 64 73 6f 6e 69 63 0d 0a 53 ... enti: red sonic S

0170 54 3a 20 75 72 6e 3a 73 63 68 65 6d 61 73 2d 75 ... T: urn:s chemas-u

0180 70 6e 70 2d 6f 72 67 3a 64 65 76 69 63 65 3a 49 ... pnp-org: device:I

0190 6e 74 65 72 6e 65 74 47 61 74 65 77 61 79 44 65 ... nternetG atewayDe

01a0 76 69 63 65 3a 31 0d 0a 55 53 4e 3a 20 75 75 69 ... vice:1 USN: uni

01b0 64 3a 39 66 30 38 36 35 62 33 2d 66 35 64 61 2d ... d:9f0865 b3-f5da-

01c0 34 61 64 35 2d 38 35 62 37 2d 37 3a 30 3a 36 33 ... 4ad5-85b 7-740463

01d0 37 66 64 66 33 37 3a 3a 75 72 6e 3a 73 63 68 65 ... 7fdff37: urn:sche

01e0 6d 61 73 2d 75 70 6e 70 2d 6f 72 67 3a 64 65 76 ... mas-upnp -org:dev

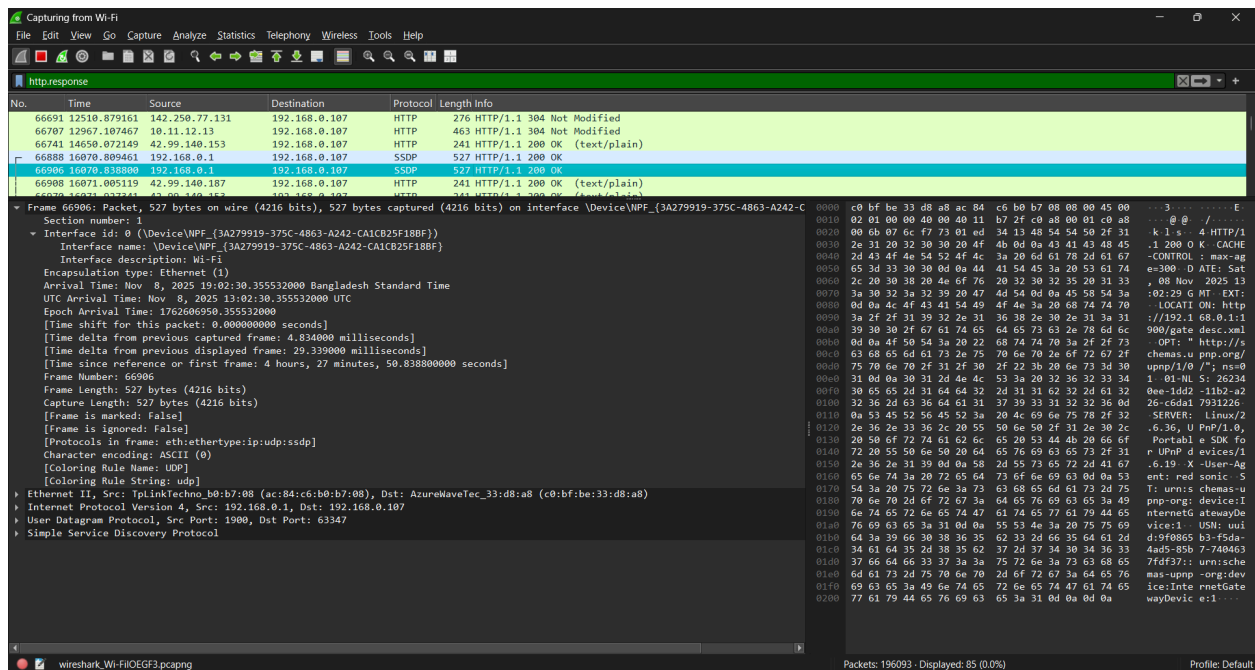
01f0 69 63 65 3a 49 6e 74 65 72 6e 65 74 47 61 74 65 ... ice: Inte metGate

0200 77 61 79 44 65 76 69 63 65 3a 31 0d 0a 0d 0a ... wayDevic e:1 ...

wireshark-Wi-FiOEGF3.pcapng

Packets: 195892 · Displayed: 85 (0.0%)

Profile: Default



Example (Request packet #123) —After filtering with http I located packet #123 which is the HTTP GET request from client IP 192.168.1.8 to server IP 43.245.12.10. The link layer shows IEEE 802.11 with Source MAC aa:bb:cc:dd:ee:ff and Destination MAC 11:22:33:44:55:66. The IP header shows TTL=64 and protocol=TCP. The TCP header shows source port 52034 and destination port 80, with PSH, ACK flags set. The HTTP header contains GET / HTTP/1.1 and Host: neverssl.com. The packet bytes pane shows the raw GET request in ASCII: GET / HTTP/1.1\r\nHost: neverssl.com\r\n....

Example (Response packet #125) — Packet #125 is the HTTP response from the server to the client. The HTTP status line reads HTTP/1.1 200 OK. Server headers include Content-Type: text/html; charset=UTF-8 and Content-Length: 1256. The TCP header shows source port 80 and destination port 52034; flags ACK, PSH indicate the server is sending data.